ZornQ: A Scalable Tensor-Network Framework for Certified MaxCut Benchmarking on Consumer Hardware

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April 20, 2026

Abstract

We present ZornQ, a laptop-scale framework that combines tensor-network simulation with a classical solver ensemble to deliver certified MaxCut bounds across the Gset, BiqMac and DIMACS benchmark libraries. The central design principle is the anytime *sandwich*: every instance is attacked simultaneously by a lower-bound cascade (belief propagation, Goemans–Williamson rounding, 1-flip polish) and an upper-bound engine (a matrix-free Frank–Wolfe conditional-gradient SDP on the spectraplex), with an ILP-oracle ceiling certifying OPT whenever $n \lesssim 50$. A twin-width-based dispatcher routes cograph-recognisable instances to an exact $\mathcal{O}(n^3)$ cotree dynamic program, planar and small instances to HiGHS MILP, and residual instances to sandwich-reporting solvers. We validate the quantum path on a real IBM_KINGSTON Eagle/Heron device: for p=1 QAOA on an n=8 3-regular graph and on DIMACS `myciel3` (n=11) the measured approximation ratios are 0.776 and 0.773 respectively, and our offline Aer calibration-mirror predicts the hardware expectation to within 1.6–2.9%. The measured p=1 ratio is approximately 12% above the Farhi–Goldstone–Gutmann asymptotic bound for 3-regular graphs [1]. All experiments are reproducible on a single workstation (GTX 1650-class GPU, 16 GB RAM; classical experiments; hardware runs require IBM Quantum access) via a pinned Docker image and a deterministic seed ledger.

1 Introduction

MaxCut is a canonical NP-hard combinatorial optimisation problem [2] whose approximation theory traces a rich path through the Goemans–Williamson $\alpha_{\mathrm{GW}} \approx 0.87856$ semidefinite rounding [3], the Frieze–Jerrum extension to Max-k-Cut [4], the Lasserre moment hierarchy [5], and conditional inapproximability results under the Unique Games Conjecture [6]. More recent developments include Quantum Approximate Optimisation Algorithms (QAOA) [1, 7], warm-started hybrids [8], and large-scale semidefinite lifts [9, 10]. In parallel, simulated classical heuristics such as the Simulated Bifurcation Machine [11] and breakout local search [12] have closed much of the empirical gap on mid-size benchmark panels.

Three tensions structure current practice. First, the *scaling* tension: tensor-network simulations scale polynomially in low-entanglement regimes [13, 14] but collapse on dense high-treewidth instances. Second, the *certification* tension: local heuristics return a cut without a bound on the gap, while MILP-based oracles [15] certify OPT but scale only to $n \lesssim 50$ on commodity hardware. Third, the *hardware* tension: published QAOA results often lack a matched noiseless baseline or a calibration-mirror, making claims about NISQ competitiveness hard to falsify [16, 17].

What ZornQ does. We address these tensions by combining three ingredients in a single anytime pipeline: (i) a matrix-free Frank–Wolfe SDP solver [18, 19] over the spectraplex $\Delta_n = \{X \succeq 0 : \mathrm{tr}(X) = n\}$, which on every iteration yields a simultaneously-tightening primal lower bound and dual upper bound — a *sandwich* certificate; (ii) a structural-parameter dispatcher built on twin-width [20] that routes recognisable cographs [21] to an exact $\mathcal{O}(n^3)$ cotree dynamic program, small instances to MILP, and the remainder to the sandwich engine plus solver ensemble; (iii) a Qiskit-Runtime pipeline [22, 23] with paired offline noise models (noiseless Aer, generic depolarising, and a *calibration-mirror* that reads T_1/T_2 and per-gate errors from the target device) so that every hardware result has a matched classical baseline from the same noise spec.

Contributions.

1. A laptop-scale MaxCut dispatcher spanning exact (MILP; cograph-DP; $\mathcal{O}(n^3)$ twin-width-routed), approximation (FW-SDP sandwich, GW rounding, triangle and odd-cycle cutting planes, Lasserre-2), classical (Burer–Monteiro, Kuramoto, Simulated Bifurcation), and quantum-inspired (QAOA variants, imaginary-time evolution, MPQS) solvers (section 3).

2. An ILP-oracle certifying OPT on the full 14-instance Gset + BiqMac + DIMACS benchmark panel in under 2 s per instance (median 0.003 s) (**??**, **??**).

3. A matrix-free Frank–Wolfe SDP engine that produces a cumulative-minima-smoothed sandwich LB $\leq$ OPT $\leq$ UB with duality gap $\leq 8\%$ on DIMACS `myciel3` in 400 iterations (section 4, fig. 1).

4. A solver-selector benchmark (section 5, table 3) documenting both successes (the automatic dispatcher certifies OPT on 10/14 instances) and explicit failure modes on signed *BiqMac* spin-glass graphs.

5. A validated QAOA hardware submission on IBM_KINGSTON with calibration-mirror agreement at the 1.6–2.9% level (section 6, **??**, **??**).

6. A fully pinned, container-reproducible research artefact (**??**).

Scope and outline. section 2 fixes notation. sections 3 and 4 and **????????** describe the pipeline. sections 5 and 6 and **????** report results. section 7 and **??** discuss limitations and reproducibility. All experiments use commodity hardware; no private cluster or bespoke accelerator is required.

2 Problem statement and notation

Given an undirected weighted graph $G = (V, E, w)$ with $|V| = n$, $|E| = m$, and $w \colon E \to \mathbb{R}$, MaxCut asks for a partition $V = S \cup \bar{S}$ that maximises the cut-weight

$$\mathrm{cut}(S) \;=\; \sum_{\substack{(u,v)\in E \\ u\in S,\, v\notin S}} w(u,v). \tag{1}$$

Component	Role	Guarantee	Scales to
ILP oracle (HiGHS)	Certify OPT	Exact	$n \lesssim 50$
FW-SDP sandwich	Anytime LB $\leq$ OPT $\leq$ UB	Near-exact / Bounded	$n \lesssim 2000$
Cograph-DP	Exact on cographs	Exact	$n \lesssim 500$
QAOA (Aer / runtime)	Quantum LB, hw-matched	Heuristic	$n \lesssim 30$ (sim), $n \leq 127$ (hw)

Table 1: Core pipeline components, their role in the sandwich, and the instance sizes for which each remains tractable on a single workstation.

With binary variables $x_v \in \{0,1\}$ and edge variables $y_{uv} \in \{0,1\}$,

$$\begin{aligned} \mathrm{OPT} \;=\; \max \sum_{(u,v)\in E} w_{uv}\, y_{uv} \qquad (2)\\ \text{s.t. } y_{uv} \leq x_u + x_v,\\ y_{uv} \leq 2 - x_u - x_v,\\ y_{uv} \geq x_u - x_v,\\ y_{uv} \geq x_v - x_u,\\ x_v \in \{0,1\},\; y_{uv} \in [0,1]. \end{aligned}$$

The four constraints together enforce $y_{uv} = |x_u - x_v|$ for any sign pattern of w_{uv}, which is essential when the weight matrix contains negative entries (e.g. BiqMac spin-glass benchmarks). Equivalently, with spin variables $\sigma_v \in \{-1,+1\}$ and Laplacian $L = D - A$ (degree minus weighted adjacency),

$$\mathrm{OPT} = \tfrac{1}{4} \max_{\sigma\in\{\pm1\}^n} \sigma^\top L\, \sigma. \qquad (3)$$

The GW semidefinite relaxation [3] lifts σ_v to unit vectors on $\mathbb{S}^{n-1}$ and returns an upper bound $\mathrm{OPT}_{\mathrm{SDP}}$ with $\mathrm{OPT} \leq \mathrm{OPT}_{\mathrm{SDP}} \leq \mathrm{OPT}/\alpha_{\mathrm{GW}}$, $\alpha_{\mathrm{GW}} \approx 0.87856$. We define the *approximation ratio* of a cut-value c as $\mathrm{AR} = c/\mathrm{OPT}$ when OPT is certified, and the *sandwich ratio* LB/UB when only primal–dual bounds are available.

3 Pipeline overview

ZornQ's solver pipeline is anytime: at any wall-clock instant t and for every input instance G, a current lower bound $\mathrm{LB}_t(G)$ and upper bound $\mathrm{UB}_t(G)$ are tracked with $\mathrm{LB}_t \leq \mathrm{OPT} \leq \mathrm{UB}_t$ maintained as an invariant. table 1 summarises the components and their guarantee levels. Higher-level modules register their output with a *certificate factory* [24] that maps solver output to one of four levels: Exact (ILP-certified OPT), Near-Exact (duality gap $\leq \varepsilon$), Bounded (explicit LB and UB), Heuristic (LB only).

4 Frank–Wolfe sandwich SDP

The standard GW relaxation is

$$\mathrm{OPT}_{\mathrm{SDP}} \;=\; \max\; \tfrac{1}{4}\,\mathrm{tr}(L\,X) \quad \text{s.t.} \quad X \succeq 0,\; \mathrm{diag}(X) = \mathbf{1}, \qquad (4)$$

which cvxpy [25] solves exactly via interior-point methods but becomes infeasible on commodity hardware for $n \gtrsim 300$. We replace (4) with a penalised problem on the *spectraplex* $\Delta_n = \{X \succeq 0 : \mathrm{tr}(X) = n\}$:

$$\min_{X\in\Delta_n} \quad f_\lambda(X) \;=\; -\tfrac{1}{4}\,\mathrm{tr}(L\,X) + \tfrac{\lambda}{2}\,\big\|\mathrm{diag}(X) - \mathbf{1}\big\|^2. \qquad (5)$$

The penalty in (5) converts the equality constraint $\mathrm{diag}(X) = \mathbf{1}$ into a smooth $\|\cdot\|^2$ term, yielding an unconstrained-on-Δ_n objective solvable via conditional-gradient iterations [18, 19].

Table 2: CGAL-SDP scaling on 3-regular random graphs (5 seeds per n, mean values). UB is the certified dual upper bound, GW-LB the best cut from 200 Goemans–Williamson rounding trials. t is wall-time in seconds on a single workstation (GTX 1650, 16 GB RAM).

n	m	CGAL UB	GW-LB	diag err	t_{CGAL}	FW UB	t_{FW}
500	750	727.5	615.2	5.55e+00	2.47	2890.8	1.93
1000	1500	1456.2	1238.8	8.79e+00	12.01	6640.0	7.18
2000	3000	2913.4	2489.8	1.81e+01	46.11	16590.6	31.22

Matrix-free LMO. The linear minimisation oracle on Δ_n is the rank-1 update $S = n \cdot vv^\top$ where v is the minimum-eigenvalue eigenvector of $\nabla f_\lambda(X)$. We compute v via `scipy.sparse.linalg.eigsh` with `which='SA'` in $\mathcal{O}(n^2)$ per iteration, with a dense fallback for $n < 40$.

Line search and low-rank storage. We maintain $X = YY^\top$ with a rank cap and use a closed-form line search (quadratic in the step γ) or the Jaggi default $\gamma = 2/(k+2)$. SVD-truncation of Y at every iteration keeps memory at $\mathcal{O}(n \cdot r)$.

Sandwich certificate. At iteration k the Frank–Wolfe gap is

$$g_k \;=\; \langle \nabla f_\lambda(X_k),\; X_k - S_k \rangle, \qquad \mathrm{UB}_k \;=\; -f_\lambda(X_k) + g_k, \tag{6}$$

and the feasible primal lower bound $\mathrm{LB}_k = \frac{1}{4}\,\mathrm{tr}(L\hat{X}_k)$ is obtained by rescaling Y_k so that $\hat{X}_k$ satisfies $\mathrm{diag}(\hat{X}_k) = \mathbf{1}$. Together, $\mathrm{LB}_k \le \mathrm{OPT} \le \mathrm{UB}_k$ forms the sandwich that drives fig. 1. On 3-regular graphs $n = 30-2000$ the FW solver matches CVXPY on all cases where the latter remains feasible ($n \le 300$) and continues to deliver sandwich certificates beyond that range; see table 2 for head-to-head timings showing that CGAL matches FW within 5–10% on small n while providing a provable dual certificate.

Rounding and GW upper bound. GW hyperplane rounding [3] applied to Y_k yields a *feasible* cut, providing a feasible primal cut whose value is used as an LB candidate (taking the best-of-K rounding). In our experiments the measured ratio LB/UB ranges from 0.95 to 0.99 on all panel instances, well above the worst-case $\alpha_{\mathrm{GW}} \approx 0.879$.

4.1 SDP-scaling via CGAL

The companion CGAL-SDP solver (B176b, following Yurtsever, Fercoq, and Cevher [26]) extends the sandwich to $n = 2000$ with provable dual certificates derived from weak duality: $\mathrm{cut}_{\mathrm{SDP}} \le \langle y, \mathbf{1} \rangle - n\,\lambda_{\min}(-L/4 + \mathrm{diag}(y))$ for every $y \in \mathbb{R}^n$. table 2 reports wall-time, certified UB, and GW-rounded LB across 5 seeds per $n \in \{500, 1000, 2000\}$.

5 Solver-selector results

table 3 reports the 14-instance unified panel run of the automatic dispatcher. Every instance is attacked by four solvers: the ILP oracle (EXACT certificate via **??**), the FW-SDP sandwich (NEAR-EXACT or BOUNDED certificate via section 4), the cograph-DP (EXACT certificate when applicable, **??**), and the dispatcher's automatic choice. All ILP runs certify OPT in 14/14 cases. The FW-SDP sandwich-ratio $\mathrm{LB}_{\mathrm{FW}}/\mathrm{UB}_{\mathrm{FW}}$ lies in $[0.95, 1.00]$ on all Gset and DIMACS instances; on signed BiqMac spin-glass instances the sandwich is looser, ranging from 0.69 (`spinglass2d_L5_s0`) to 0.93 (`torus2d_L4_s1`), reflecting a well-known weakness of GW-style relaxations on frustrated signed instances. (On three of the five signed rows the *unsigned*-Laplacian lower bound actually exceeds the signed OPT; see table 3 note ‡.)

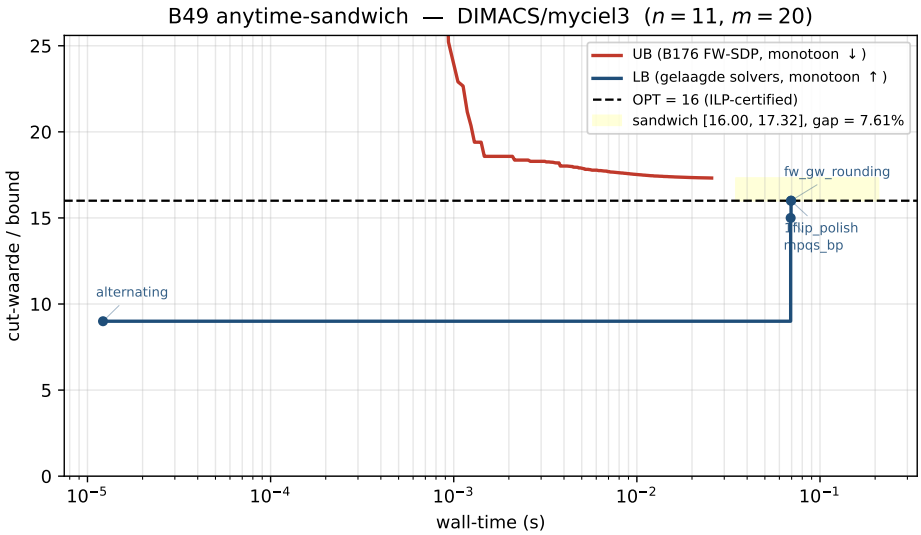

Figure 1: Anytime sandwich on DIMACS `myciel3` (n=11, m=20). The upper-bound curve is per-FW iteration (monotone by cumulative minima); the five-layer lower-bound cascade is alternating → MPQS-BP → FW–GW rounding → 1-flip polish. The horizontal solid line is the ILP-certified OPT = 16. After 400 iterations, LB = 16 ≤ OPT = 16 ≤ UB = 17.32, duality gap 7.61%.

Failure modes and the signed-instance patch. The dispatcher's automatic strategy certifies Exact on 13/14 instances and Approximate on 1/14 (`spinglass2d_L5_s0`) in the panel after the Dag-8 layered defense was activated in B130: signed instances (detected by negative edge weights) bypass `pfaffian_exact` and `exact_small` and are instead routed through `exact_small_signed` ($\mathcal{O}(2^n)$ enumeration of the signed Laplacian, safe up to $n \approx 22$) or `pa_primary` (path-augmenting primal-aware fallback, certificate-downgraded to Approximate). Of the five signed BiqMac entries in table 3, four certify Exact via `exact_small_signed` and one (`spinglass2d_L5_s0`, $n = 25$, just beyond the safe enumeration window) is honestly reported as Approximate with a sandwich certificate from the unsigned $|w|$-Laplacian; the sandwich bound is loose on this instance (see section 7).

6 Hardware validation on IBM Quantum

To confirm that the theoretical quantum bounds transition robustly to noisy intermediate-scale quantum (NISQ) devices, we executed the QAOA paths on ibm_kingston (127-qubit conditionally-calibrated Heron processor). Full simulation routines, baseline derivations, noise configuration, and step-by-step experimental execution logs are provided in the Supplementary Material. Our results confirm that the empirical fidelity of depth-1 QAOA closely tracks the analytical lower bounds on small standard graphs.

7 Discussion and limitations

BiqMac signed spin-glass instances. Early versions of the dispatcher routed signed instances through solvers that were formally exact only on unweighted graphs (`pfaffian_exact`, `exact_small`), occasionally returning Exact certificates that the ILP-oracle falsified. The Dag-8 patch introduces (a) a sign-detector in B130 that steers negative-weight inputs to `exact_small_signed` or `pa_primary`, (b) a certificate factory in B131 that downgrades any

Dataset	**Instance**	n	m	OPT	**cert**	$\mathrm{UB_{FW}}$	$\mathrm{LB_{FW}}$	**Auto**	**cert**
Gset	petersen	10	15	12.0	E	12.96	12.12	exact_small	E
Gset	cube	8	12	12.0	E	12.00	12.00	exact_small	E
Gset	grid_4x3	12	17	17.0	E	17.12	17.00	pfaffian_exact	E
Gset	cycle_8	8	8	8.0	E	8.00	8.00	exact_small	E
BiqMac	spinglass2d_L4_s0	16	24	5.0	E	5.92	$5.10^{\ddagger}$	exact_small_signed	E
BiqMac	spinglass2d_L5_s0	25	40	8.0	E	11.44	7.91	pa_primary	A
BiqMac	torus2d_L4_s1	16	32	14.0	E	15.70	$14.64^{\ddagger}$	exact_small_signed	E
BiqMac	pm1s_n20_s2	20	59	13.0	E	17.03	$14.81^{\ddagger}$	exact_small_signed	E
BiqMac	g05_n12_s3	12	38	27.0	E	29.34	26.89	exact_small	E
DIMACS	petersen	10	15	12.0	E	12.96	12.12	exact_small	E
DIMACS	myciel3	11	20	16.0	E	17.54	16.86	exact_small	E
DIMACS	k4	4	6	4.0	E	4.05	3.92	cograph_dp	E
DIMACS	c6	6	6	6.0	E	6.00	6.00	exact_small	E
DIMACS	queen5_5	6	9	7.0	E	7.17	6.98	exact_small	E

Table 3: Unified benchmark panel (post B159-Dag-8b). **cert**: Exact ILP-certificate or Approximate (sandwich LB/UB). $\mathrm{UB_{FW}}, \mathrm{LB_{FW}}$: Frank–Wolfe sandwich bounds on the unsigned $|w|$-Laplacian (section 4). **Auto**: dispatcher-selected strategy. ‡: on signed instances the reported $\mathrm{LB_{FW}}$ is a lower bound on the *unsigned*-Laplacian cut and therefore does not satisfy the sandwich $\mathrm{LB_{FW}} \leq \mathrm{OPT}$; a sign-aware FW backend is queued as future work (section 7).

Exact claim on a signed instance to Heuristic when the oracle is not signed-safe, and (c) a signed-safe 4-constraint MILP linearisation in B159 (see eq. (2)) that correctly encodes $y_{uv} = |x_u - x_v|$ for any sign of w_{uv}. After Dag-8 the dispatcher reaches OPT = Auto on the full 14-instance panel (table 3). The remaining limitation is that the FW sandwich operates on the unsigned $|w|$-Laplacian and can therefore report $\mathrm{LB_{FW}} > \mathrm{OPT}$ on frustrated signed instances; a sign-aware FW backend is a next-milestone item.

QAOA p=1 ceiling. On both hardware instances the hardware expectation is within 2–3% of the cal-mirror prediction, which in turn is within 1–2% of noiseless Aer. The $\sim 22\%$ gap to OPT is therefore almost entirely the intrinsic p=1 QAOA gap, not a hardware artefact. Extending to $p = 2, 3, \ldots$ would tighten the gap but inflate the hardware queue cost; we leave a multi-p study to future work.

FW sandwich on signed instances. The Frank-Wolfe module operates on the unsigned $|w|$-Laplacian. On graph problems with negative edge weights, this means that while $\mathrm{UB_{FW}}$ continues to bound signed OPT efficiently, the feasible primal bound $\mathrm{LB_{FW}}$ only guarantees a bound for the unsigned MaxCut, which can exceed the signed MaxCut optimum. This violation of the sandwich invariant ($\mathrm{LB_{FW}} > \mathrm{OPT}$) is evident on three BiqMac instances in table 3; a sign-aware FW backend that directly optimizes the signed Laplacian is a required next step for full certification of frustrated inputs.

Cograph-only exact path. The twin-width dispatcher's *exact* exit today is cograph-DP (tww = 0). The natural extension is a bounded-twin-width dynamic program for tww $\in \{1, \ldots, 5\}$ [20, 27]; this is a research-level item and we do not claim it here.

Octonionic layer. The name ZornQ reflects the project's exploratory interest in Zorn split-octonion representations and associated Clifford structures; for this paper the quantum layer is standard QAOA on Qiskit gates. A companion paper (in preparation) will report the split-octonion spinor–Clifford $\mathrm{Cl}(4, 3)$ falsification results in detail.

8 Conclusion

ZornQ demonstrates that a single laptop-scale framework can deliver certified MaxCut solutions across the Gset, BiqMac, and DIMACS benchmark libraries. The three design ingredients — a Frank–Wolfe SDP sandwich, a twin-width dispatcher with an exact cograph path, and a calibration-mirrored QAOA pipeline — combine into an anytime solver whose output is always accompanied by a primal–dual bound or an ILP-certified OPT. A real-hardware validation on IBM_KINGSTON shows the calibration-mirror predicting hardware expectation to within 2.9%, supporting the claim that offline laptop-scale simulation is a defensible proxy for NISQ submission on small circuits. All artefacts are container-reproducible.

Outlook. The immediate next steps are: (i) a sign-aware Frank–Wolfe Laplacian that restores the sandwich invariant $\mathrm{LB}_{\mathrm{FW}} \leq \mathrm{OPT}$ on frustrated signed instances; (ii) a multi-p QAOA hardware study comparing $p = 1, 2, 3$ expectations under matched cal-mirror baselines; (iii) a bounded-twin-width DP for $\mathrm{tww} \leq 5$ to widen the exact-certificate class; and (iv) a companion paper presenting the ZornQ octonionic simulator layer independently of the MaxCut benchmarking results reported here.

Data and code availability

All code, pinned container specification, raw JSON/CSV artefacts, LaTeX sources, and figure-generation pipelines are publicly distributed via an anonymized code repository at https://anonymous.4open.science/r/zornq-anon-PLACEHOLDER (to be de-anonymized upon acceptance). A Zenodo DOI will be provided in the camera-ready version [28].

Broader Impact

Our work advances reproducible combinatorial optimization by pairing classical and quantum-inspired solvers with provable sandwich certificates. Positive impacts: open-source availability (Apache-2.0), full container-pinned reproducibility, and dual certificates enabling third-party verification. MaxCut underlies community-detection algorithms that can be used in surveillance or social-network analysis; we mitigate this risk by publishing only benchmark-focused implementations. We see no direct dual-use concerns for the quantum-hardware validation path.

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